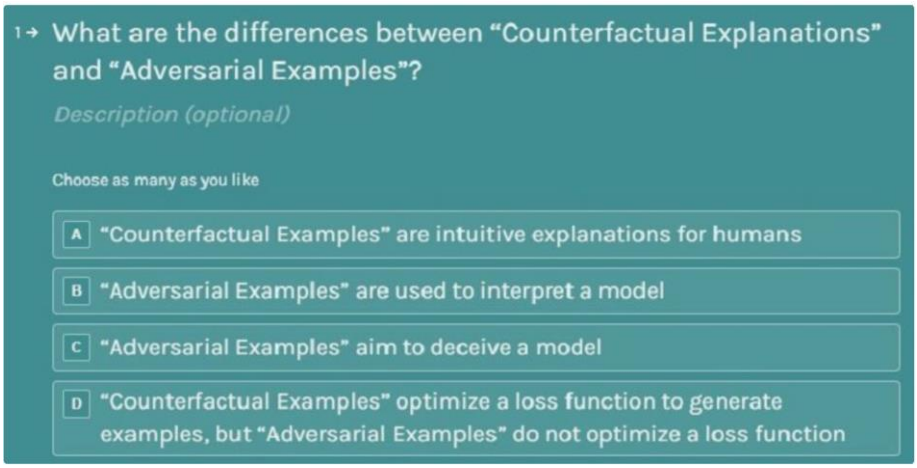
XAI questions and answers (tricky answers are sometimes with reference)

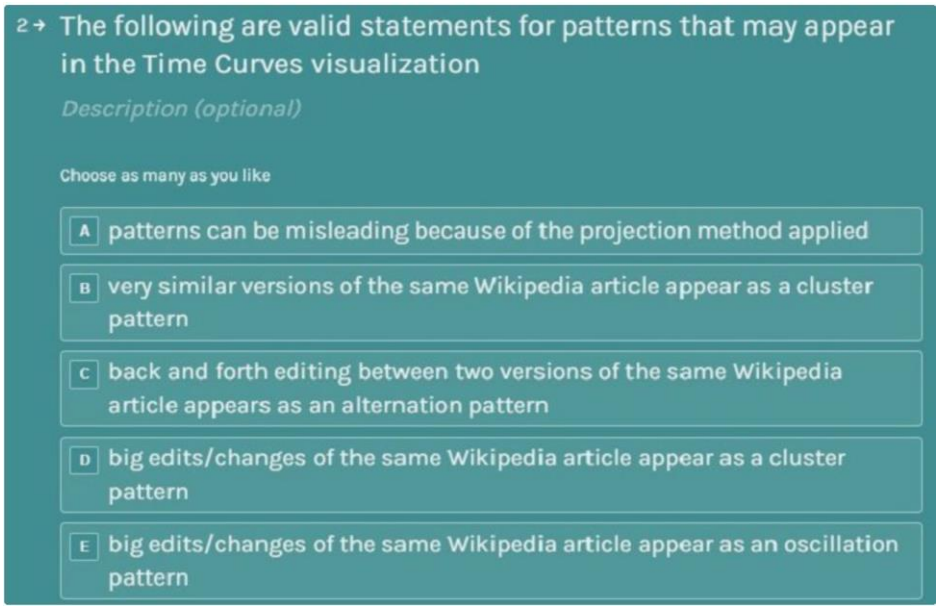
Legend: ln 5 lecture notes 5

1. What are the diff



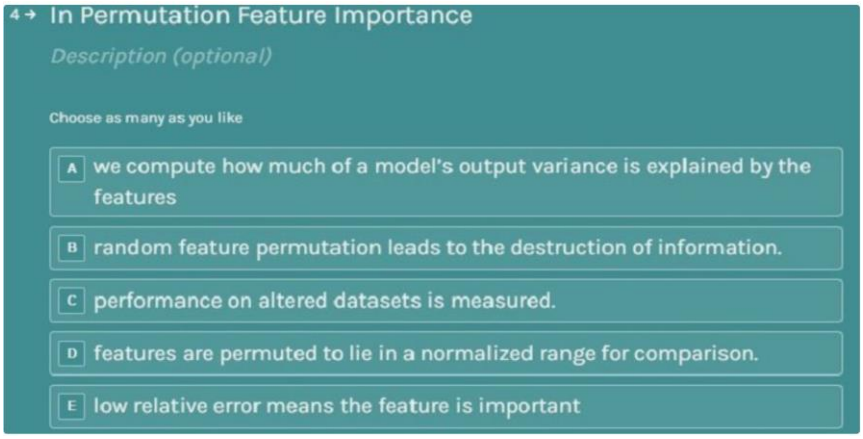
   a. yes
   b. No: ln 5
   c. Yes: ln 5
   d. No, because adversarial examples also optimize a loss function. (REF: https://christophm.github.io/interpretable-ml-book/adversarial.html)

2. The following are ln3



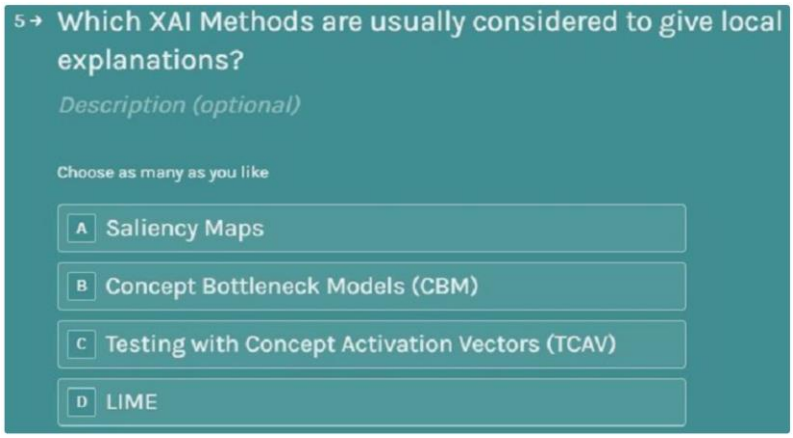
   a. unlikely
   b. yes
   c. yes
   d. no big edits/changes are shown as an oscillation
   e. yes

3. clear
4. In permutation Feature importance. ln 5



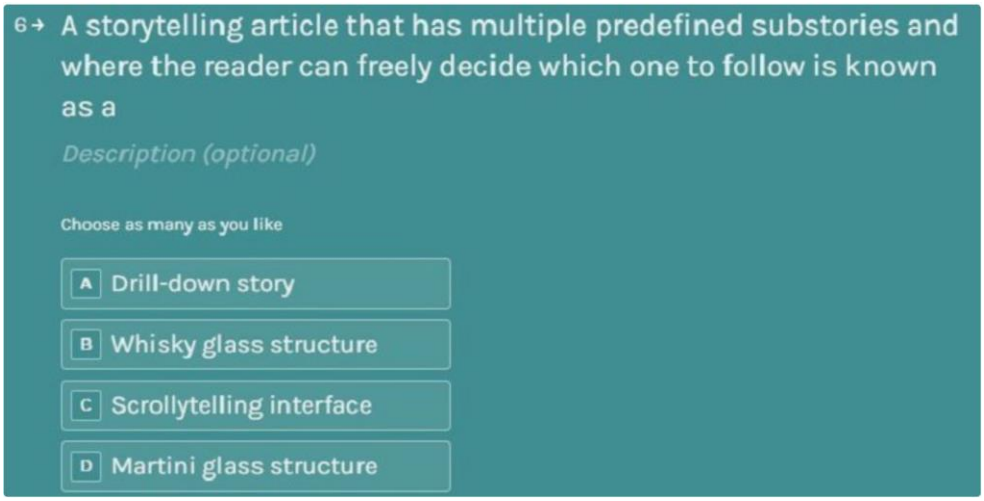
a. no error of model is used, not variance
b. yes
c. yes
d. no
e. no high relative error means the feature is important

5. Which XAI Methods are usually



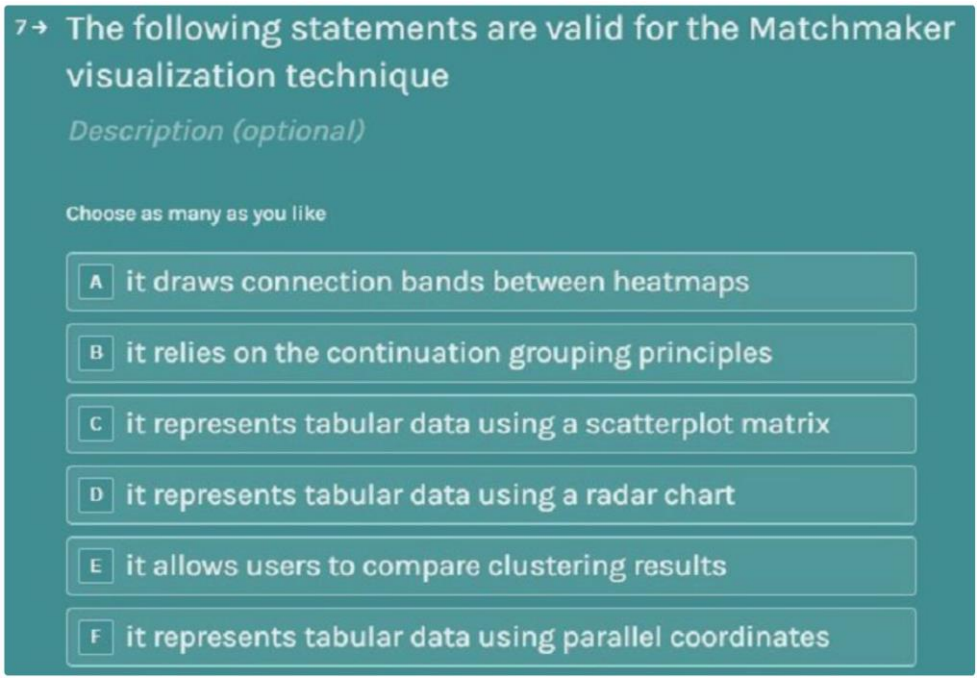
a. yes
b. yes
c. no global
d. yes

6. A storytelling



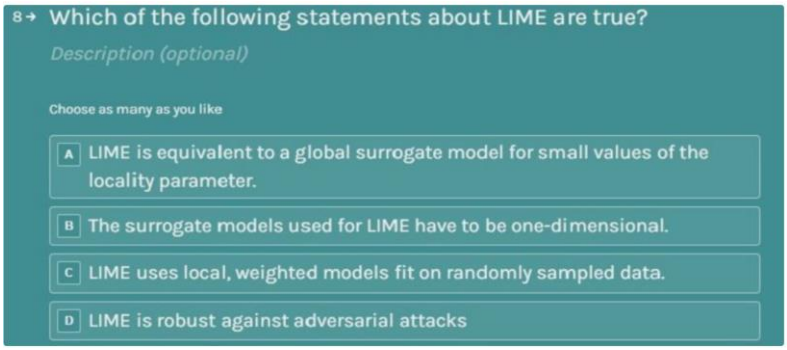
a. yes

7. The following statements are valid https://www.youtube.com/watch?v=vi-G3LqHFZA



   a. yes
   b. yes
   c. no
   d. no
   e. yes
   f. yes

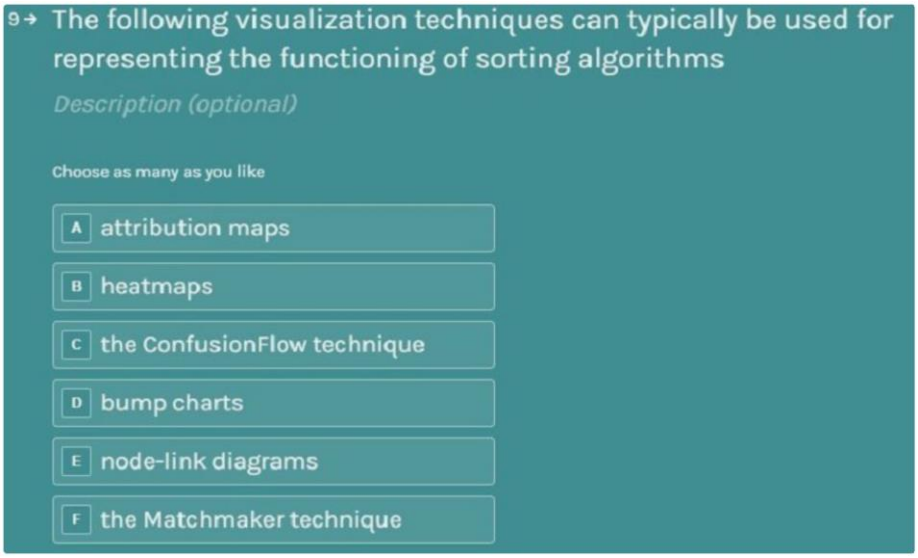
8. Which of the following statements about LIME are true?



   a. LIME is equivalent to a global surrogate model for small values of the locality parameter. I would say opposite.
   b. The surrogate models used for LIME have to be one-dimensional. The guy in lecture is showing it on a one-dimensional model, But is this general the case?
   c. LIME uses local, weighted models fit on randomly sampled data. No, data are not randomly sampled, LIME currently uses an exponential smoothing kernel to define the neighborhood. reference Molnar book
   d. LIME is robust against adversarial attacks. No, opposite.

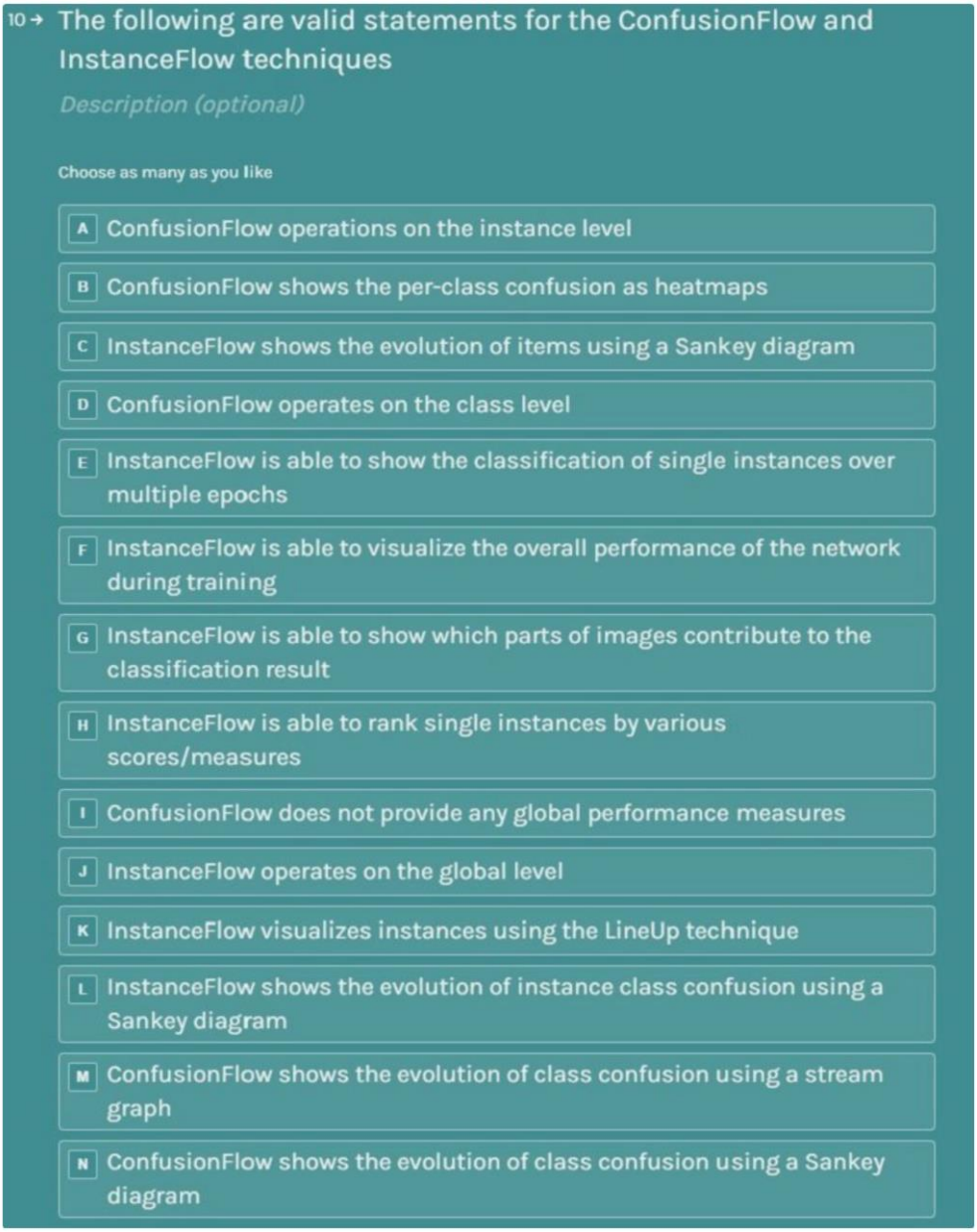
According to Marc Streit at least b is incorrect.

9. The following



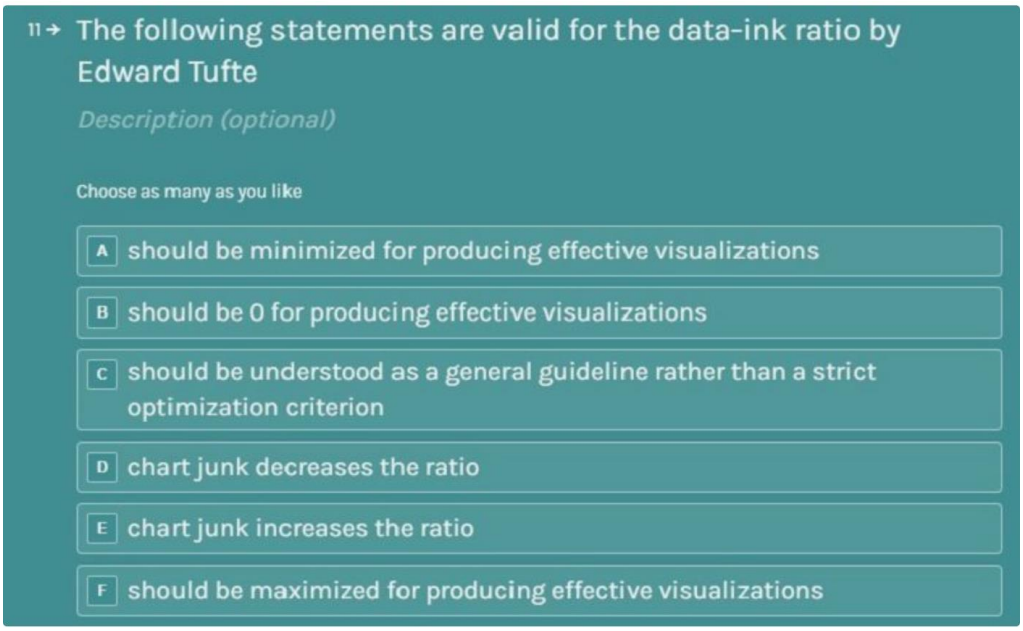
a. no: attribution map: An attribution map signals pixels in the input image highly relevant to the downstream tasks or hidden representations, and therefore highlights the “attention” of a network over a specific task. In other words, the model space can be thought as a space defined on top of attribution maps, where the affinity between points or networks is evaluated using the distance between their produced attribution maps, which again, requires no supervision and can be computed really fast.
b. yes
c. no, Confusion Flow visualizes NN Confusion across epochs
d. yes, ln 2
e. yes, googling: sorting algorithm explained with nodes results in a few hits. Seems that topological sorting is often explained with node-link diagrams.
f. no: The Caleydo Matchmaker technique allows users to split multidimensional datasets into subgroups, cluster them separately and analyze relations between the resulting clusters

10. The following are



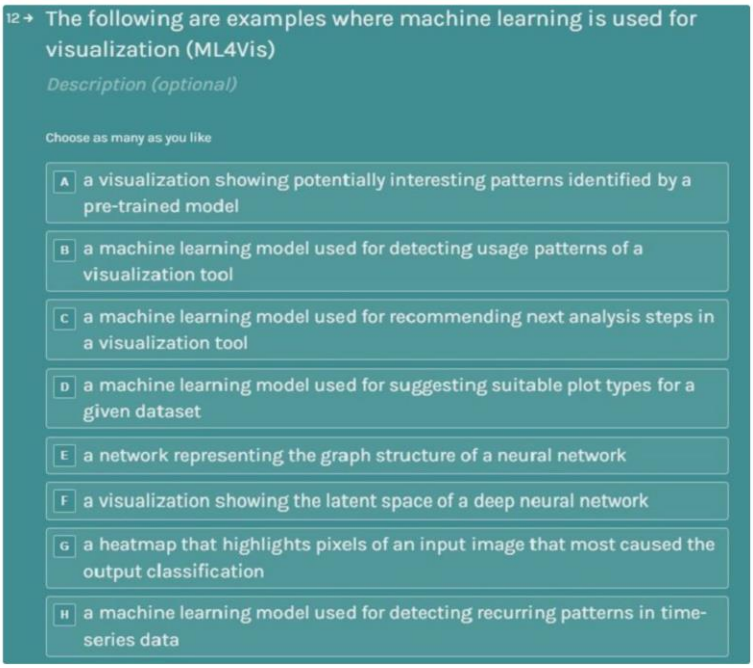
a. no, on class level
b. yes
c. yes
d. yes
e. yes
f. no
g. no, gradcam is doing this
h. no
i. no
j. no, on instance level
k. yes; reference https://www.youtube.com/watch?v=EAyvPtf84ns 5:20
l. yes
m. no
n. no, just line charts and heat maps

11. The following statements



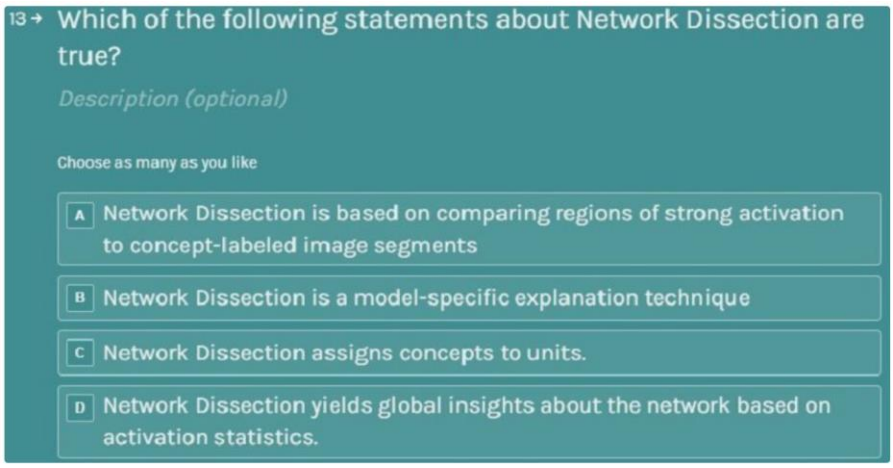
a. no
b. no. should be 1
c. yes
d. yes
e. no
f. yes

12. The following are examples



correct: b,c,d, h

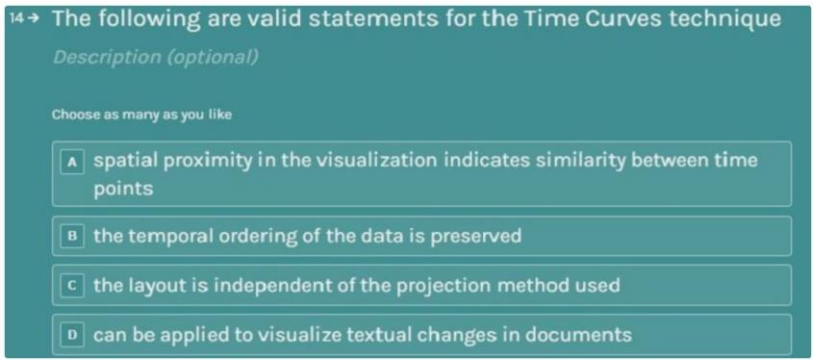
13. Which of the statements about Network Dissection are true?



Network Dissection is **an interpretability method for CNNs that evaluates the alignment between individual hidden units and a set of visual semantic concepts**. By identifying the best alignments, units are given human interpretable labels across a range of objects, parts, scenes, textures, materials, and colors.

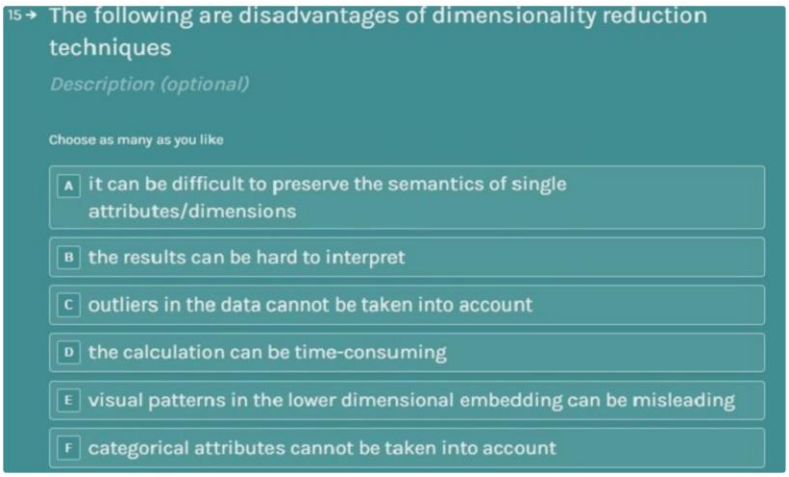
a. yes
b. yes
c. no; units are matched to concepts
d. no can be local or global depending on usage

14. The following are valid statements for the Time curve technique



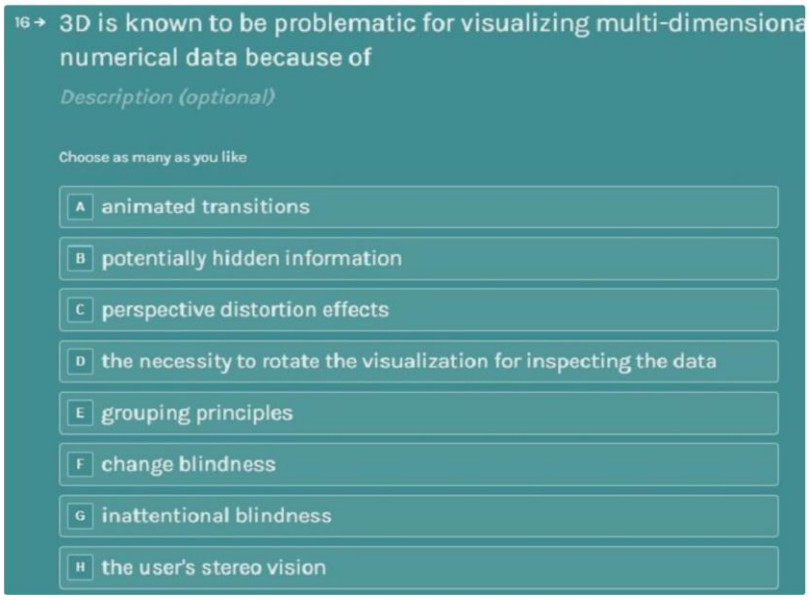
a. in caption of Fig 1 in https://hal.inria.fr/hal-01205821/document they write Fig. 1. The time curve principle: a) a timeline is folded into itself in such a way that similar time points end up being close to each other; b) Example: a time curve showing the evolution of a Wikipedia article. In my opinion not really clearly formulated. They should have written: Time points with similar context end up close to each other. Streit formulated the answer similar to the figure caption. So I will tick it in the exam and complain if they mark it as false.
b. yes
c. no
d. yes

15. The following are disadvantages



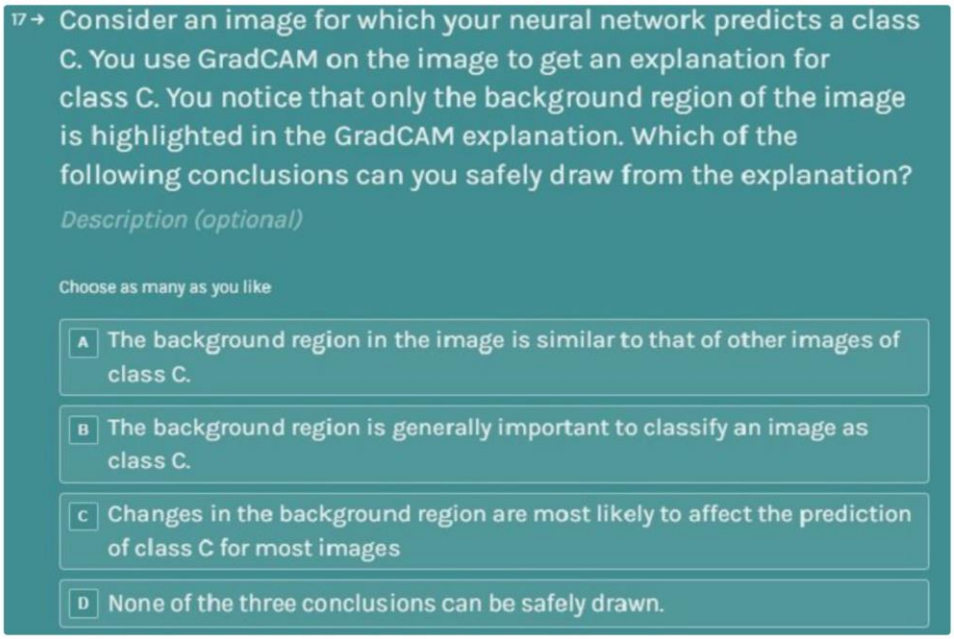
a. yes
b. yes
c. no
d. yes, there is visual analytics exam 22.2.21 where not ticking this resulted in loss of points. (t-SNE is slow)
e. yes
f. no

16. 3D is known to be



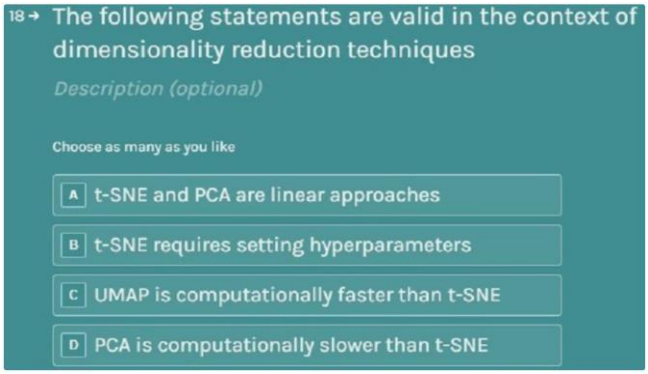
a. no
b. probably no, because its not hidden in a way that you cant find it at all....its just occluded and can be seen through rotation and zooming. IDK what the correct answer is though.
c. yes
d. yes
e. no
f. no (problematic for animations)
g. no
h. no

17. Consider an image for which



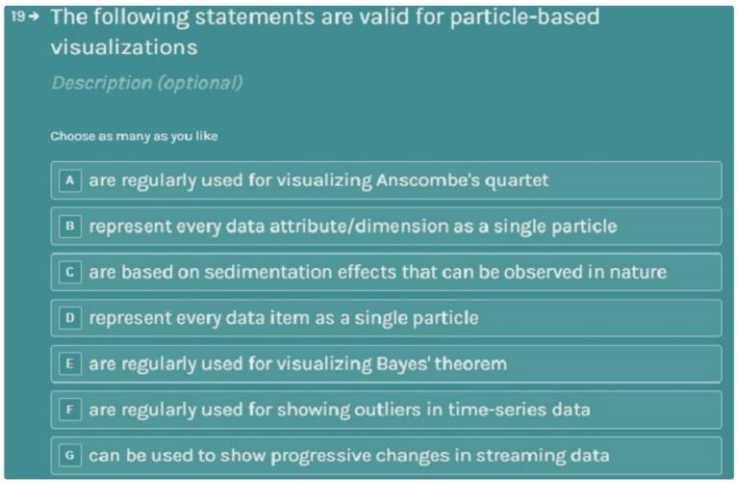
a. probably no
b. probably yes
c. yes
d. no

18. The following statements



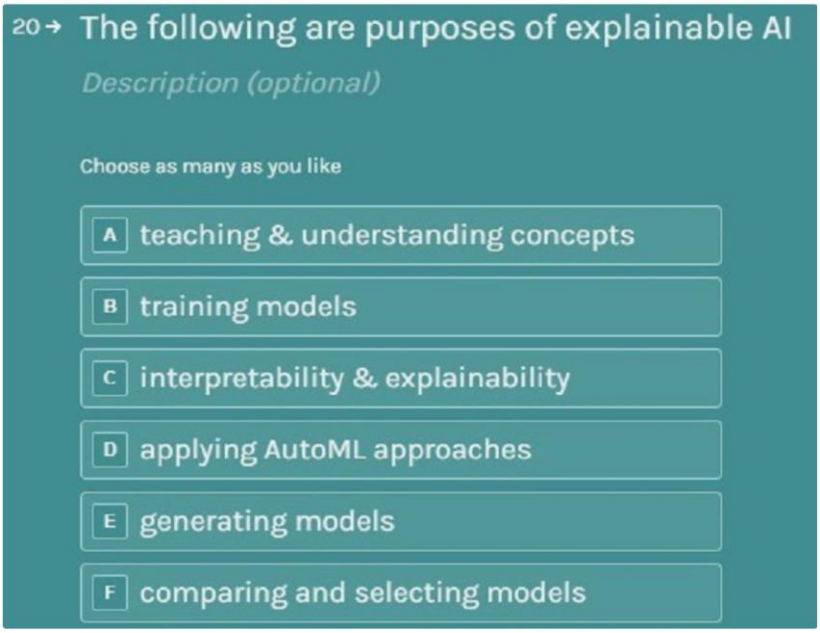
a. no PCA and MDS are linear
b. yes
c. yes
d. no, opposite

19. The following statements are



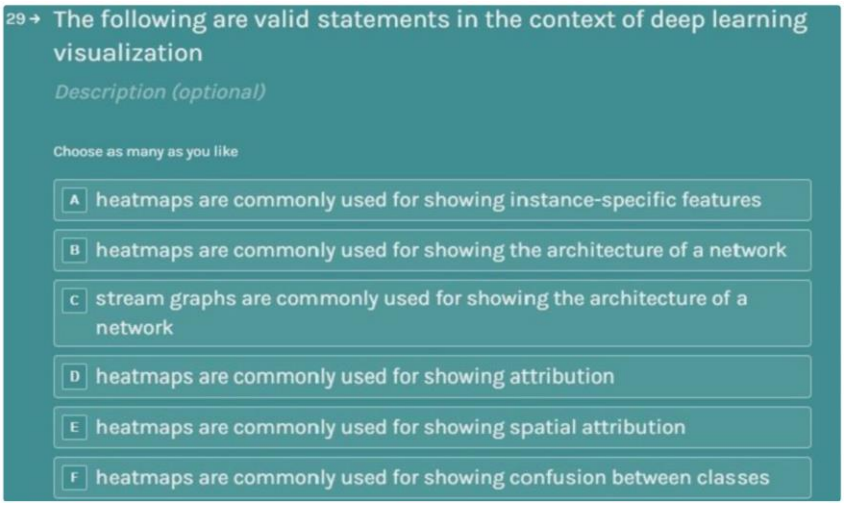
a. probably no. But tricky again. Watch lecture 2 time ~1h9min. There he is saying particle based visualization can be compared to snow falling down or sedimentation. In Anscombe's quartet we have a x,y tuple. However, the lecture notes say: idea: Show every data point as a single particle ; Asking Marc Streit he answered that answer a is incorrect, although not saying if my reasoning is correct.
b. no every data item is a single particle
c. yes ln 2
d. yes
e. yes, he is saying this at lecture 2 time~ 1h9min
f. no
g. probably yes

20. The following are purposes



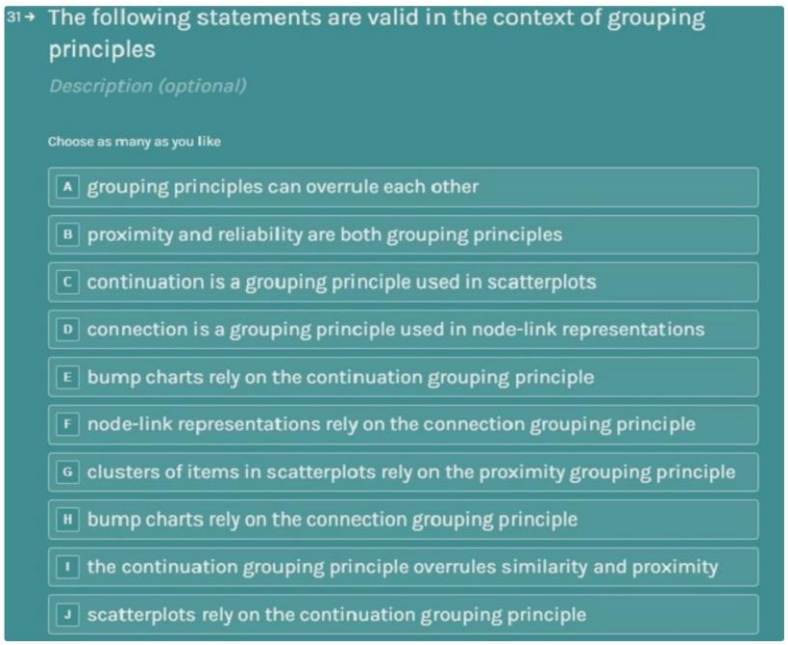
a. yes
b. no
c. yes
d. no
e. no
f. yes

21. clear
22. clear
27. clear
28. clear
29. The following are valid statements



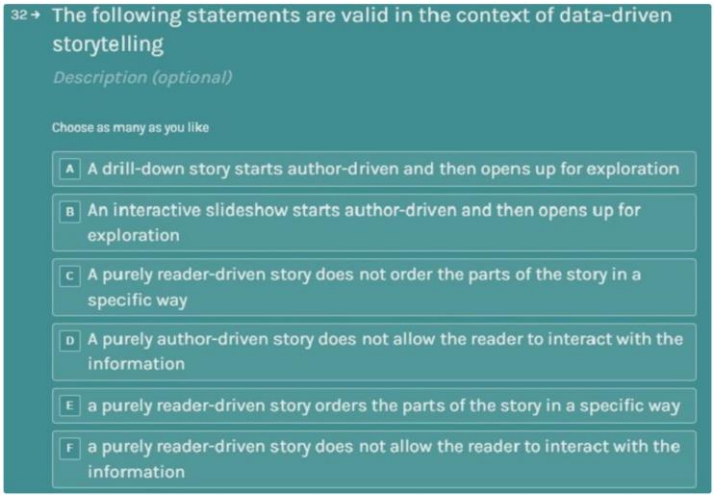
a. yes
b. no
c. no https://en.wikipedia.org/wiki/Streamgraph
d. yes, see Q9
e. no, usually things are cut out
f. yes, in confusion matrices

30. clear
31. The following statements are valid



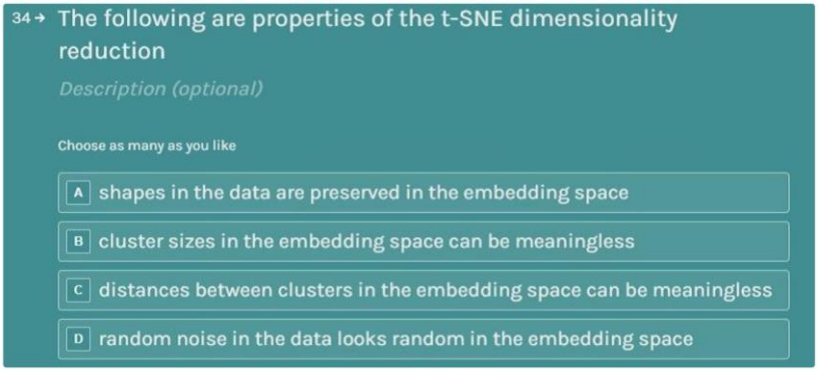
a. yes; connection is a very strong grouping principle and can overrule others
b. reliability no
c. no
d. yes
e. yes
f. yes
g. yes
h. yes
i. no
j. no

32. The following statements are valid



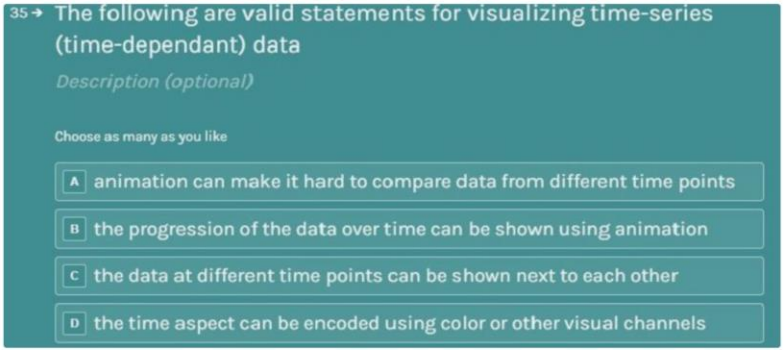
a. no
b. no
c. yes
d. yes
e. no
f. no

33. clear
34. The following are properties



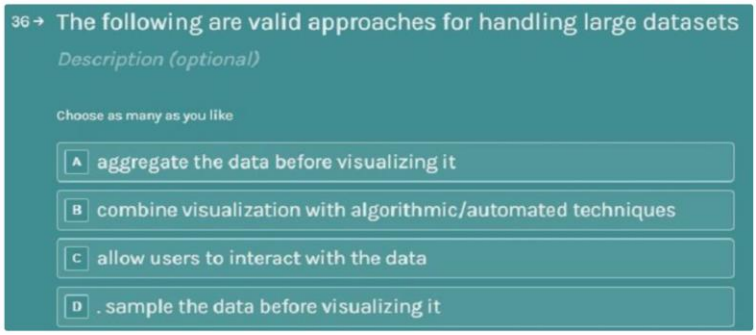
a. no
b. yes
c. yes
d. no

35. The following are



a. yes
b. yes
c. yes; Answer verified my Marc Streit: Yes, that's correct. Juxtaposing the visualizations for the different time points can actually be very effective in many scenarios because it allows easy comparison (without the need to interpret an animation).
d. yes

36. The



a. yes
b. yes
c. yes
d. yes https://www.computerweekly.com/de/definition/Data-Sampling-Datenauswahl

JKU JOHANNES KEPLER UNIVERSITY LINZ

Ways To Deal With Large Datasets

Temporal Partitioning
- Navigation: Pan, Rotate, Zoom
- Geometric vs. Semantic Zooming

Spatial Partitioning
- Multiple Coordinated Views
- Overview + Detail

Aggregation & Filtering

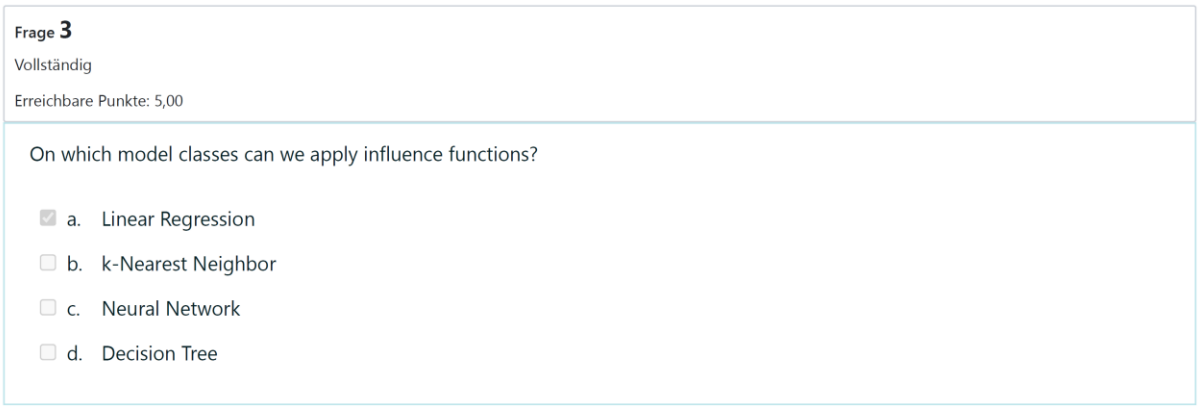
NEW Questions from retry exam 23.2.2023

Frage **2**
Vollständig
Erreichbare Punkte: 5,00

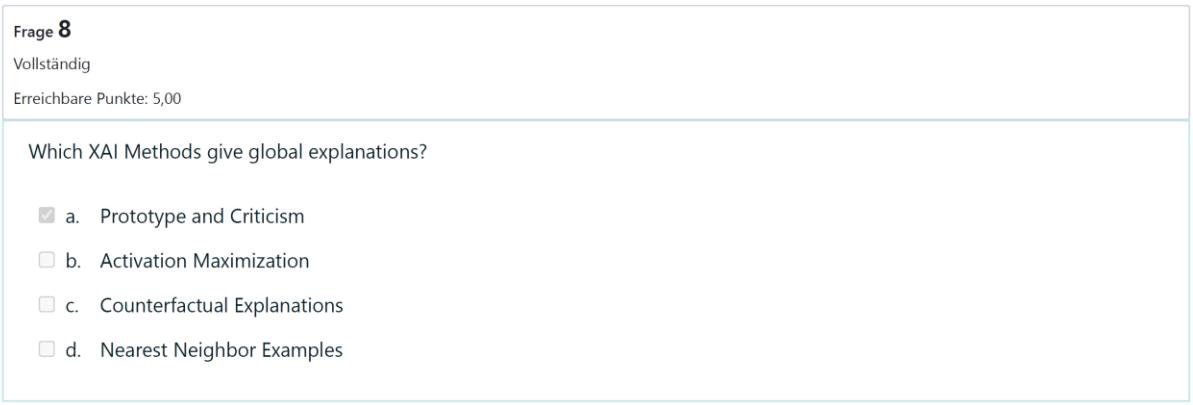
What is true about Influential Instances?

- [ ] a. Cannot give local explanations.
- [ ] b. Cannot give global explanations.
- [ ] c. Can give global explanations.
- [x] d. Can give local explanations.

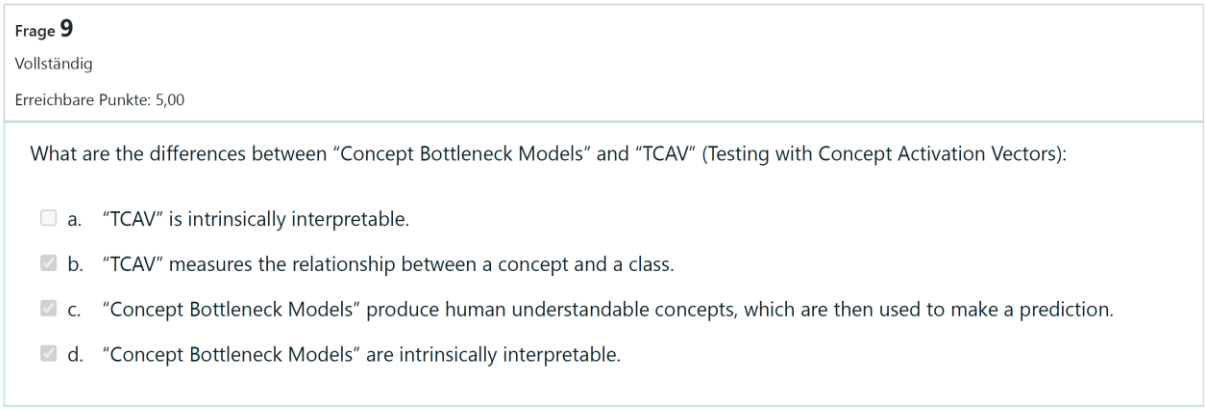
correct: c,d



correct: a,c



correct: a,b



a. TCAV is posthoc, therefore not intrinsically interpretable
b. yes
c. yes
d. yes

3.2 Taxonomy of Interpretability Methods

Methods for machine learning interpretability can be classified according to various criteria.

Intrinsic or post hoc? This criteria distinguishes whether interpretability is achieved by restricting the complexity of the machine learning model (intrinsic) or by applying methods that analyze the model after training (post hoc). Intrinsic interpretability refers to machine learning models that are considered interpretable due to their simple structure, such as short decision trees or sparse linear models. Post hoc interpretability refers to the application of interpretation methods after model training. Permutation feature importance is, for example, a post hoc interpretation method. Post hoc methods can also be applied to intrinsically interpretable models. For example, permutation feature importance can be computed for decision trees. The organization of the chapters in this book is determined by the distinction between intrinsically interpretable models and post hoc (and model-agnostic) interpretation methods.